M13 Batch 5 — Full Set

M13.6 Equipment & Furnishings and M13.8 Instruments · PDF pages 322–448

Cabin equipment and emergency requirements, instrument classification, the atmosphere, pressure measuring devices, pitot static systems and their faults.

265 questions · Level 3: 93 · Level 2: 128 · Level 1: 44

● marks a question built on a fact your instructor highlighted or underlined in the manual (224 of 265). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Equipment and furnishings: requirements and cabin systems

PDF pages 332–336 · 28 questions

LEVEL 1 — RECALL

1. For which minimum period must a required public address system remain powered after the shutdown or failure of all engines and auxiliary power units ●
   a. At least 3 minutes
   b. At least 5 minutes
   c. At least 10 minutes

2. On which technology is the Cabin Intercommunication Data System based ●
   a. An analogue amplifier chain
   b. A microprocessor
   c. A hardwired relay logic

3. How many subsystems make up the Advanced Cabin Entertainment/Service System ●
   a. Seven
   b. Five
   c. Four

4. Which subsystem broadcasts flight deck announcements, flight attendant announcements and boarding music to the cabin ●
   a. The Passenger Address System
   b. The Cabin Interphone System
   c. The Passenger Entertainment System

LEVEL 2 — UNDERSTAND AND EXPLAIN

5. A restraining means is fitted to a megaphone. Which paragraph of the design specification sets the inertia forces it must withstand?
   a. CS 25.561, Emergency Landing Conditions
   b. CS 25.1423, Public Address System
   c. CS 25.1421, Megaphones

6. An announcement is made on a required public address system. Where in the aeroplane must it be intelligible? ●
   a. At all passenger seats, lavatories and flight crew rest compartments
   b. At all passenger seats, lavatories and cabin crew seats and work stations
   c. At all passenger seats, galleys and flight crew seats and work stations

7. Which other system must a required public address system be able to work independently of ●
   a. Any required cabin crew member call system
   b. Any required passenger entertainment system
   c. Any required crew member interphone system

8. Which emergency exits must have a microphone readily accessible to the seated cabin crew member ●
   a. Required floor-level exits with an adjacent passenger seat
   b. Required floor-level exits with an adjacent cabin crew seat
   c. Required overwing exits with an adjacent cabin crew seat

9. Which three things does the Cabin Intercommunication Data System do to the cabin systems it serves ●
   a. Controls, tests and monitors them
   b. Controls, records and displays them
   c. Tests, records and announces them

10. Which users are the cabin systems of the Cabin Intercommunication Data System dedicated to ●
    a. Cabin attendant and ground service use
    b. Cabin attendant and passenger use
    c. Flight crew and ground crew use

11. Where does the audio of the Passenger Address System go ●
    a. To the cabin speakers and the passenger headphones
    b. To the cabin speakers and the attendant handsets
    c. To the passenger headphones and the flight deck speakers

12. Video, announcements and recorded audio all have to reach a seat. Which subsystem controls and distributes them to it? ●
    a. The Passenger Services System
    b. The Passenger Entertainment System
    c. The Cabin Interphone System

13. Smoke is picked up by a lavatory smoke detector. Which cabin subsystem takes that signal and gives the annunciation? ●
    a. The Cabin Lighting System
    b. The Cabin Interphone System
    c. The Passenger Services System

14. The flight deck needs to speak to the load cargo master. Which subsystem carries that call? ●
    a. The Passenger Address System
    b. The Cabin Configuration System
    c. The Cabin Interphone System

15. Which lights are controlled by the Cabin Lighting System ●
    a. The ceiling and wash lights
    b. The passenger reading lights
    c. The passenger call lights

16. Besides the ceiling, wash and night illumination lights, which further lights does the Cabin Lighting System control ●
    a. The crew rest and reading lights
    b. The lavatory call and seat row lights
    c. The lavatory and crew rest lights

17. Which benefit does the Cabin Configuration System give an airline ●

a. Entertainment audio selectable at every seat
b. Interior changes with minimal hardware changes
c. Fault reporting of the whole cabin to the flight deck

18. How is an airplane interior configuration change accomplished through the Cabin Configuration System ●

a. By reprogramming each passenger service unit
b. By exchanging the configuration pins of the cabin wiring
c. By modifying the ACESS configuration database

19. A new cabin configuration has to be prepared before it is loaded. Which equipment creates the ACESS configuration database? ●

a. On-ground personal computer software
b. On-board central maintenance computer
c. On-board cabin configuration module

20. Which cabin items can be arranged through the ACESS configuration database ●

a. The seat recline limits and the galley loads
b. The PA areas and the chime mute zones
c. The chime tone and the ceiling light brightness

21. Which test of the ACESS Test System runs on-line and continuously ●

a. The built-in test equipment test
b. The built-in test
c. The programming test

22. A major fault is found by the ACESS Test System. Besides the Cabin Configuration/Test Module, where is it reported? ●

a. To the airplane configuration system and the ground station
b. To the EICAS/EFIS interface units and the maintenance computers
c. To the passenger service units and the attendant call panels

LEVEL 3 — APPLY AND REASON

23. The same power source that feeds the public address system also feeds other loads that stay powered when every other source is inoperative. How must those other loads be treated? ●

a. Those other loads are shed for the whole powering duration
b. Those other loads are taken into account in the powering duration
c. Those other loads are transferred to the crew interphone supply

24. A microphone has been left out of its stowage and unused at a cabin crew station. What must the public address system do in that condition? ●

a. Become inoperative until that microphone is stowed
b. Mute the loudspeakers in that cabin zone
c. Remain fully operative in the cabin

25. Two required floor-level exits lie close together and each has an adjacent cabin crew member seat. When may one microphone serve both exits? ●

a. When the two crew member seats face in the same direction
b. When the two exits are on the same side of the fuselage
c. When the seated crew members can talk unassisted

26. Two operators want the cabin of the same aeroplane arranged to suit their own service. What does the CIDS design philosophy give them? ●

- **a.** Flexibility gained by exchanging the cabin wiring looms
- **b.** Flexibility without complex and time expensive hardware changes
- **c.** A fixed configuration set once by the airframe manufacturer

27. Which two items are the special features of the integrated cabin entertainment and service system rather than its subsystems ●

- **a.** The Passenger Services System and the Passenger Address System
- **b.** The Cabin Configuration System and the ACESS Test System
- **c.** The Cabin Lighting System and the Cabin Interphone System

28. A reading light and a night illumination light both fail in the same cabin section. Why are two different subsystems involved? ●

- **a.** Reading lights belong to the lighting system and night illumination to the services system
- **b.** Reading lights belong to the services system and night illumination to the address system
- **c.** Reading lights belong to the services system and night illumination to the lighting system

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Answers — Equipment and furnishings: requirements and cabin systems

1. **c** The specification sets a time duration of at least 10 minutes. (L1, PDF p. 332)

2. **b** The CIDS is described as a microprocessor based system. (L1, PDF p. 334)

3. **b** Five subsystems are listed for the ACESS. (L1, PDF p. 336)

4. **a** Those announcements and the boarding music are broadcast through the Passenger Address System. (L1, PDF p. 336)

5. **a** The megaphone paragraph itself refers the inertia forces out to CS 25.561. (L2, PDF p. 332)

6. **b** Intelligibility is demanded at the passenger seats, the lavatories and the cabin crew seats and work stations. (L2, PDF p. 332)

7. **c** Independence is demanded specifically from any required crew member interphone system. (L2, PDF p. 332)

8. **b** The requirement is written for required floor-level exits that have an adjacent cabin crew member seat. (L2, PDF p. 332)

9. **a** Control, test and monitor are the three functions printed for the CIDS. (L2, PDF p. 334)

10. **b** The cabin systems concerned are those dedicated to cabin attendant and passenger use. (L2, PDF p. 334)

11. **a** PA audio is fed to the cabin speakers and to the passenger headphones. (L2, PDF p. 336)

12. **b** All three are routed to the seat through the Passenger Entertainment System. (L2, PDF p. 336)

13. **c** The interface to the lavatory smoke detector for smoke detection annunciation is a feature of the Passenger Services System. (L2, PDF p. 336)

14. **c** Flight deck to load cargo master is one of the calls made through the Cabin Interphone System. (L2, PDF p. 336)

15. **a** The ceiling and wash lights are in the Cabin Lighting System list. (L2, PDF p. 336)

16. **c** The lavatory and crew rest lights complete the Cabin Lighting System list. (L2, PDF p. 336)

17. **b** The CCS lets the airline change the interior configuration with minimal hardware changes. (L2, PDF p. 336)

18. **c** Configuration changes are made by modifying the ACESS configuration database. (L2, PDF p. 336)

19. **a** The Airplane Configuration System that creates the database is an on-ground PC software. (L2, PDF p. 336)

20. **b** PA areas and chime mute zones are printed in the list of configurable items. (L2, PDF p. 336)

21. **b** The built-in test is the on-line continuous one, while the BITE is off-line. (L2, PDF p. 336)

22. **b** Major faults go on to the EIUs and the CMCs as well as to the CCTM. (L2, PDF p. 336)

23. **b** The duration must be achieved while considering the other loads left on the same source. (L3, PDF p. 332)

24. **c** No unused, un-stowed microphone may render the system inoperative, so the system has to remain fully operative. (L3, PDF p. 332)

25. **c** One microphone may cover several exits only if their proximity allows unassisted verbal communication between the seated cabin crew members. (L3, PDF p. 332)

26. **b** The CIDS is designed for high flexibility so that complex and time expensive hardware changes are not needed. (L3, PDF p. 334)

27. **b** The CCS and the ATS are named as the two special features added to the five subsystems. (L3, PDF p. 336)

28. **c** Reading lights sit under the Passenger Services System while night illumination sits under the Cabin Lighting System. (L3, PDF p. 336)

Network integrated cabin equipment

PDF pages 338–338 · 4 questions

LEVEL 1 — RECALL

29. Which kind of system is the nice system ●

- **a.** A modular IP based cabin management system
- **b.** A centralised IP based flight management system
- **c.** A modular analogue cabin lighting system

LEVEL 2 — UNDERSTAND AND EXPLAIN

30. For which aircraft is the nice cabin management system offered ●

- **a.** Long range freighter aircraft
- **b.** VIP and government aircraft
- **c.** Regional commuter aircraft

31. Which cabin functions can the crew and the passengers reach through the nice system ●

- **a.** Illumination, cabin pressure, audio/video and internet access
- **b.** Window-shades, galley power, audio/video and seat heating
- **c.** Illumination, window-shades, audio/video and internet access

32. Further functions are to be added to an existing nice installation. Which protocol makes the system easy to expand? ●

- **a.** The spanning tree protocol
- **b.** The ARINC 429 protocol
- **c.** The standard TCP/IP protocol

Answers — Network integrated cabin equipment

29. **a** The nice system is a modular, IP based cabin management system. (L1, PDF p. 338)

30. **b** The system is offered for VIP and government aircraft. (L2, PDF p. 338)

31. **c** Illumination, window-shades, audio/video and internet access are the cabin functions printed. (L2, PDF p. 338)

32. **c** Expandability comes from the use of the standard TCP/IP protocol. (L2, PDF p. 338)

Instrument classification and cockpit layout

PDF pages 347–361 · 17 questions

LEVEL 1 — RECALL

33. To which two subgroups can every instrument of the classical cockpit be assigned ●

a. Airframe monitoring and aircraft monitoring
b. Flight monitoring and aircraft monitoring
c. Flight monitoring and engine monitoring

34. To which group does the artificial horizon indicator ADI belong ●

a. Airframe monitoring
b. Aircraft monitoring
c. Flight monitoring

35. A panel layout is described as Basic T. Which four instruments form that T? ●

a. Airspeed indicator, attitude indicator, altimeter and compass
b. Attitude indicator, altimeter, compass and turn coordinator
c. Airspeed indicator, attitude indicator, altimeter and vertical speed indicator

LEVEL 2 — UNDERSTAND AND EXPLAIN

36. Of these three flight monitoring instruments, the one that is also provided as a standby device in the modern cockpit is ●

a. The compass
b. The machmeter
c. The variometer

37. The group of instruments which the standby set of a modern cockpit duplicates is ●

a. flight monitoring
b. engine monitoring
c. airframe monitoring

38. An exhaust gas temperature indicator is fitted in a classical cockpit. Which group and subdivision does it belong to? ●

a. Aircraft monitoring, as an engine monitoring device
b. Flight monitoring, as a pressure monitoring device
c. Aircraft monitoring, as an airframe monitoring device

39. Into which two kinds of devices is aircraft monitoring divided ●

a. Airframe monitoring devices and standby devices
b. Airframe monitoring devices and engine monitoring devices
c. Flight monitoring devices and engine monitoring devices

40. Why do the aircraft monitoring systems of the modern cockpit carry different names such as EICAS or ECAM ●

a. Because the name depends on the aircraft type
b. Because the name depends on the display size
c. Because the name depends on the flight phase

41. Which task do EICAS and ECAM perform in the modern cockpit ●

- **a.** They perform aircraft monitoring
- **b.** They perform maintenance data recording
- **c.** They perform flight monitoring

42. Why is the CMS counted as part of aircraft monitoring ●

- **a.** Because it delivers information to the maintenance personnel
- **b.** Because it delivers information to the standby instruments
- **c.** Because it delivers information to the flight crew

43. The electronic displays of a glass cockpit are lost in flight. Which speed and attitude information remains available from the standby set? ●

- **a.** Mach number from the standby machmeter and attitude from the standby horizon
- **b.** Airspeed from the standby airspeed indicator and heading from the standby horizon
- **c.** Airspeed from the standby airspeed indicator and attitude from the standby horizon

44. A fuel pressure indication changes during flight. Which value is the instrument responding to?

- **a.** Pressure of air
- **b.** Earth's magnetism
- **c.** Pressure of fluids

45. A pilot reads his magnetic heading and a VOR course deviation from one round instrument. Which indicator is he using? ●

- **a.** The HSI
- **b.** The ADI
- **c.** The variometer

46. A glass cockpit aircraft has no separate round flight instruments. Which statement describes the arrangement of the flight parameters? ●

- **a.** They are laid out on the PFD and correspond to the Basic T configuration
- **b.** They are laid out on the PFD in a configuration that replaces the Basic T
- **c.** They are laid out on the navigation display in Basic T configuration

LEVEL 3 — APPLY AND REASON

47. An aircraft presents its altimeter, airspeed indicator and artificial horizon indicator on cockpit screens. To which system of the modern cockpit are these instruments assigned? ●

- **a.** EFIS (Electronic Flight Instrument System)
- **b.** EICAS (Engine Indicating and Crew Alerting System)
- **c.** CMS (Central Maintenance System)

48. An engineer needs aircraft system information for troubleshooting rather than an indication meant for the crew. Which aircraft monitoring system delivers it? ●

- **a.** The EFIS (Electronic Flight Instrument System)
- **b.** The CMS (Central Maintenance System)
- **c.** The ECAM (Electronic Centralized Aircraft Monitoring)

49. An instrument is to be fitted in the middle at the top of a Basic T panel and must also display the indication from the flight director system. Which indicator is required? ●

- **a.** The ADI
- **b.** The turn coordinator
- **c.** The HSI

Answers — Instrument classification and cockpit layout

33. **b** Classical cockpit instruments divide into flight monitoring instruments and aircraft monitoring instruments. (L1, PDF p. 347)

34. **c** The ADI is listed among the flight monitoring instruments. (L1, PDF p. 347)

35. **a** The four instruments arranged in the T shape are the airspeed indicator, attitude indicator, altimeter and compass; turn coordination and vertical speed sit below them. (L1, PDF p. 361)

36. **a** The standby devices are the altimeter, the airspeed indicator, the horizon and the compass; the machmeter and the variometer appear only in the flight monitoring list. (L2, PDF p. 347)

37. **a** Altimeter, airspeed indicator, horizon and compass are all listed as flight monitoring instruments. (L2, PDF p. 347)

38. **a** Exhaust gas temperature indicators are named as engine monitoring devices under aircraft monitoring. (L2, PDF p. 347)

39. **b** Aircraft monitoring covers airframe monitoring devices and engine monitoring devices. (L2, PDF p. 347)

40. **a** Aircraft monitoring instruments carry different names depending on the aircraft type. (L2, PDF p. 347)

41. **a** EICAS and ECAM are the names given to the aircraft monitoring instruments. (L2, PDF p. 347)

42. **a** The CMS is used for aircraft monitoring because it delivers information to maintenance personnel. (L2, PDF p. 347)

43. **c** The standby set contains an airspeed indicator and a horizon, but no machmeter. (L2, PDF p. 347)

44. **c** Fluid pressure is listed separately from air pressure as a value that operates instruments. (L2, PDF p. 347)

45. **a** The HSI is the instrument that combines compass and radio navigation information. (L2, PDF p. 361)

46. **a** The PFD keeps the Basic T arrangement rather than abandoning it. (L2, PDF p. 361)

47. **a** All three named instruments are listed as flight monitoring instruments, and flight monitoring instruments go to the EFIS. (L3, PDF p. 347)

48. **b** The CMS is the aircraft monitoring system that delivers information to the maintenance personnel. (L3, PDF p. 347)

49. **a** The middle top position of the Basic T carries the attitude indicator, and the ADI is the attitude indicator that also shows the flight director indication. (L3, PDF p. 361)

The atmosphere, ISA and terminology

PDF pages 365–375 · 26 questions

LEVEL 1 — RECALL

50. What is the proportion of nitrogen in dry, clean air directly above the surface of the earth ●

- **a.** 78%
- **b.** 1%
- **c.** 21%

51. What is the proportion of oxygen in dry, clean air directly above the surface of the earth ●

- **a.** 78%
- **b.** 21%
- **c.** 11%

52. What does HUD stand for ●

- **a.** Heading Up Display
- **b.** Head Up Display
- **c.** Head Unit Display

53. The transition level is the specified last usable flight level. By what minimum margin must it lie above the 5000 ft MSL level ●

- **a.** 1000 ft
- **b.** 500 ft
- **c.** 2000 ft

LEVEL 2 — UNDERSTAND AND EXPLAIN

54. How does the temperature behave with increasing height in the troposphere ●

- **a.** It increases by 0.65 °C every 100 metres
- **b.** It decreases by 6.5 °C every 100 metres
- **c.** It decreases by 0.65 °C every 100 metres

55. The air data measurement also called atmospheric pressure and the one also called ram pressure are ●

- **a.** the static pressure and the dynamic pressure respectively
- **b.** the static pressure and the total pressure respectively
- **c.** the dynamic pressure and the static pressure respectively

56. The two measured pressures alone are not enough for two of the air data indications. Those two indications are ●

- **a.** barometric altitude and true airspeed
- **b.** true airspeed and static air temperature
- **c.** airspeed and vertical speed

57. How do static pressure, temperature and density behave as altitude increases ●

- **a.** All three become lower
- **b.** Only temperature becomes lower
- **c.** All three become higher

58. The true airspeed is to be indicated as well as the pressures already measured. Which value must be applied as a corrective? ●

- **a.** Density
- **b.** Humidity
- **c.** Temperature

59. A pneumatic instrument is calibrated in the workshop. Which values form the basis of that calibration?

- **a.** The actual airfield values at the time of installation
- **b.** The tropopause values of the standard atmosphere
- **c.** The standard atmosphere values at mean sea level

60. Why has the international standard atmosphere been established

- **a.** To predict the weather conditions along a planned route
- **b.** To set the pressure scale of the altimeter before every flight
- **c.** To compare flight performance under greatly differing conditions

61. How does the HUD present the PFD data ●

- **a.** So that the pilot can see it only when the autopilot is engaged
- **b.** So that the pilot can see it when he looks down at the centre pedestal
- **c.** So that the pilot can see it when he looks through the windshield

62. A crew selects QNH on both altimeters. In which phases of flight is this setting used? ●

- **a.** During landing and take-off, up to transition altitude and below transition level
- **b.** During landing and take-off, above transition altitude and above transition level
- **c.** During cruise, above transition altitude and above transition level

LEVEL 3 — APPLY AND REASON

63. A crew flies at a level that lies inside the troposphere over the equator but above it over the poles. Which printed thicknesses account for this? ●

- **a.** About 16.8 km at the equator and about 11 km at the poles
- **b.** About 16.8 km at the equator and about 8.5 km at the poles
- **c.** About 8.5 km at the equator and about 16.8 km at the poles

64. An aircraft climbs and its static pressure falls. The indication a mechanical air data instrument derives from that falling pressure is ●

- **a.** the indicated airspeed
- **b.** the static air temperature
- **c.** the barometric altitude

65. An airspeed indicator works on the difference between the total and the static pressure. That difference is which of the two air data measurements required for flight monitoring ●

- **a.** the static pressure
- **b.** the barometric altitude
- **c.** the dynamic pressure

66. In cruise above the transition altitude all pilots set 1013.2 hPa on the altimeter. Which standard atmosphere value is this setting taken from? ●

- **a.** The pressure at the transition level
- **b.** The pressure at mean sea level
- **c.** The pressure at the tropopause

67. How does the air temperature of the standard atmosphere behave with increasing height directly above the tropopause

- **a.** It begins to rise with height
- **b.** It stops falling and stays constant
- **c.** It continues to fall at the same rate

68. The air data instruments of an aircraft are electrical indicators fed from a central computing unit. In this fit the mechanical conversion of the pressure into electricity takes place

- **a.** in the pitot and static probes
- **b.** in the air data calculators
- **c.** in the indicating instrument itself

69. One screen in the cockpit carries all the air data information and also every Basic T parameter. This screen is ●

- **a.** the standby horizon
- **b.** the PFD (Primary Flight Display)
- **c.** the HUD (Head Up Display)

70. Two altimeters on the same parked aircraft are read at the airfield, one showing zero and the other showing the airfield elevation. Which settings are selected?

- **a.** QFE on the first and QNH on the second
- **b.** QFE on the first and QNE on the second
- **c.** QNH on the first and QFE on the second

71. An aircraft climbs through the transition altitude after take-off. Which change of altimeter setting is required? ●

- **a.** From STD (QNE) to QNH
- **b.** From QFE to QNH
- **c.** From QNH to STD (QNE)

72. An aircraft descends through the transition level on the approach. Which change of altimeter setting is required? ●

- **a.** From QNH to QFF
- **b.** From STD (QNE) to QNH
- **c.** From QNH to STD (QNE)

73. The low range radio altimeter is unserviceable and the crew needs the height above the airfield. Which setting gives it on the barometric altimeter, and under which condition?

- **a.** QFE, in as far as the aircraft stays below the transition altitude
- **b.** QNH, in as far as there is no change in air pressure
- **c.** QFE, in as far as there is no change in air pressure

74. An aircraft in cruise above the transition altitude keeps QNH set instead of the standard pressure. What is no longer ensured

- **a.** the measurement of the height above the ground
- **b.** the safe vertical separation from other aircraft
- **c.** the indication of the airfield elevation

75. An aircraft cruises at 4000 ft above mean sea level in an area with a transition altitude of 5000 ft. Which altimeter setting must be used?

- **a.** QNH
- **b.** QFF
- **c.** QNE

Answers — The atmosphere, ISA and terminology

50. **a** Nitrogen makes up 78% of dry clean air at the surface. (L1, PDF p. 365)

51. **b** Oxygen makes up 21% of dry clean air at the surface. (L1, PDF p. 365)

52. **b** HUD is short for Head Up Display. (L1, PDF p. 373)

53. **a** A margin of at least 1000 ft above the 5000 ft MSL level is required. (L1, PDF p. 375)

54. **c** The troposphere lapse rate is 0.65 °C per 100 m of climb. (L2, PDF p. 365)

55. **a** Static pressure is also called atmospheric pressure, and dynamic pressure is also called ram pressure. (L2, PDF p. 367)

56. **b** Temperature as a corrective value makes true airspeed and static air temperature available. (L2, PDF p. 367)

57. **a** All three listed values fall with increasing altitude. (L2, PDF p. 367)

58. **c** Temperature is the corrective value that makes TAS and SAT available. (L2, PDF p. 367)

59. **c** Calibration is referred to the ISA mean sea level values. (L2, PDF p. 367)

60. **c** The ISA exists so performance figures can be compared on a common basis. (L2, PDF p. 369)

61. **c** The HUD projects the PFD information into the pilot's forward view. (L2, PDF p. 373)

62. **a** QNH belongs to the low level phases, below the transition altitude or level. (L2, PDF p. 375)

63. **b** The troposphere reaches much higher at the equator than at the poles, so the same level can be inside it in one place and above it in the other. (L3, PDF p. 365)

64. **c** Static pressure falls as altitude increases, and it is the static pressure that a mechanical instrument uses for barometric altitude. (L3, PDF p. 367)

65. **c** Dynamic pressure is the difference between total and static pressure, so of the two required measurements it is the dynamic pressure the indicator works on. (L3, PDF p. 367)

66. **b** The standard atmosphere fixes the sea level pressure at 1013,25 hPa, which is the pressure the crew sets for flight levels. (L3, PDF p. 369)

67. **b** Identical temperatures at both levels show the lapse rate has stopped at the tropopause. (L3, PDF p. 369)

68. **b** Where the instruments are electrical, the mechanical conversion from pressure to electricity happens in the air data calculators and not in the instrument. (L3, PDF p. 371)

69. **b** The PFD is the screen that holds all the air data information, and it shows every one of the Basic T parameters. (L3, PDF p. 373)

70. **a** With QFE the barometric altimeter reads zero on the airfield, while with QNH it reads the elevation there. (L3, PDF p. 375)

71. **c** The transition altitude is the height at which the altimeter is changed from QNH to the standard setting. (L3, PDF p. 375)

72. **b** The transition level is the flight level at which the altimeter is changed from the standard setting back to QNH. (L3, PDF p. 375)

73. **c** Height is normally taken from the low range radio altimeter, and the barometric altimeter on QFE gives it only as long as the air pressure does not change. (L3, PDF p. 375)

74. **b** The common 1013.2 hPa setting in cruise is what ensures safe vertical separation, so an aircraft left on QNH no longer has it. (L3, PDF p. 375)

75. **a** Standard setting only applies above the transition altitude, so below it the altimeter stays on QNH. (L3, PDF p. 375)

Pressure measuring devices

PDF pages 377–393 · 67 questions

LEVEL 1 — RECALL

76. An operator prepares its fleet for RVSM operations. What does RVSM do to the separation minima? ●

a. The halving of vertical safety distances at higher flight altitudes
b. The doubling of vertical safety distances at higher flight altitudes
c. The halving of lateral safety distances at higher flight altitudes

77. What is RVSM airspace ●

a. Airspace above FL 285 with 500 ft vertical separation
b. Airspace above FL 290 with 2000 ft vertical separation
c. Airspace above FL 290 with 1000 ft vertical separation

78. What does the abbreviation IAS stand for ●

a. Indicated airspeed
b. Inertial airspeed
c. Increased airspeed

79. Which airspeed is displayed in aircraft equipped with air data computers and glass cockpits ●

a. The true airspeed
b. The equivalent airspeed
c. The calibrated airspeed

80. For which purpose is the true airspeed needed ●

a. For design data purposes
b. For navigation purposes
c. For engine trim purposes

81. What is the function of a Bourdon tube ●

a. It converts pressure into a proportional temperature signal
b. It converts pressure into a proportional electrical signal
c. It converts vibration into a proportional electrical signal

82. Which cross-sectional shape does the curved tube of a Bourdon transducer have ●

a. An oval shape
b. A square shape
c. A round shape

83. Which stroke do diaphragm capsules supply as the measurable value ●

a. 14 to 15 mm
b. 4 to 5 mm
c. 0.4 to 0.5 mm

LEVEL 2 — UNDERSTAND AND EXPLAIN

84. What is the purpose of reducing the vertical separation minimum ●

a. To reduce the workload of the air data computers
b. To accommodate more aircraft in the same airspace
c. To allow aircraft to climb above the tropopause

85. Why was the separation above FL 285 formerly increased

- **a.** Lack of measurement precision and reduced altimeter sensitivity at high altitude
- **b.** Lack of autopilot authority and reduced control response at high altitude
- **c.** Lack of radar coverage and reduced transponder range at high altitude

86. Besides the air data computers, the altimeters and the autopilot in altitude hold mode, which further item is RVSM-sensitive? ●

- **a.** The radio altimeter antennas and their surroundings
- **b.** The transponder antenna and its surroundings
- **c.** The structure and surroundings of the pitot-static sensors

87. Which transponder requirement applies to an aircraft flying in RVSM airspace? ●

- **a.** It must transmit altitude and be connected to the radio altimeter
- **b.** It must transmit heading and be connected to the altitude measurement unit
- **c.** It must transmit altitude and be connected to the altitude measurement unit

88. Which operational advantage is gained by an RVSM approved aircraft ●

- **a.** It can use the increased number of flight levels to fly above the tropopause
- **b.** It can use a reduced number of flight levels to save maintenance costs
- **c.** It can use the increased number of flight levels to save time and fuel

89. A workshop is being prepared for the maintenance of RVSM approved aircraft. Which of the mandatory training subjects concerns the accuracy of the workshop's own measuring instruments?

- **a.** Improved precision of test equipment
- **b.** The configuration of air data systems
- **c.** Improved altimeter precision

90. A fleet of six aircraft is to be cleared for RVSM. Who must apply for the Airworthiness Approval and for what? ●

- **a.** The aircraft operator, for the whole fleet at once
- **b.** The aircraft operator, for each individual aircraft
- **c.** The aircraft manufacturer, for each individual aircraft

91. What does the requirement for RVSM Airworthiness Approval mean for each individual aircraft ●

- **a.** Modification according to required service bulletins and the altitude monitoring programme
- **b.** Modification according to required service bulletins and a full static system overhaul
- **c.** Replacement of the air data computers and the altitude monitoring programme

92. The RVSM training covers surface waviness in the critical area. Which causes of that waviness are named? ●

- **a.** Rivets, oxygen discharge discs and corrosion pits
- **b.** Fasteners, oxygen discharge discs and paint drops
- **c.** Fasteners, static dischargers and paint drops

93. Which detail of the air data computer must the mandatory RVSM training cover ●

- **a.** Its cooling airflow
- **b.** Its part numbers
- **c.** Its software loading procedure

94. Which approval makes the aircraft eligible for the RVSM test flight ●

- **a.** The RVSM Operational Approval
- **b.** The MNPS Approval
- **c.** The RVSM Airworthiness Approval

95. The altitude monitoring programme and the test flight have both been completed successfully. Which approval does the operator now receive? ●

a. The MNPS Approval
b. The RVSM Airworthiness Approval
c. The RVSM Operational Approval

96. An airspeed indicator with bellows is connected to the pitot and static systems. Which pressure does its indication directly follow? ●

a. The total pressure as the sum of dynamic and ambient pressure
b. The static pressure as the difference between ambient and total pressure
c. The dynamic pressure as the difference between total and static pressure

97. Under which conditions does an airspeed indicator with bellows show the real speed of the aircraft ●

a. At cruise altitude with standard conditions
b. At mean sea level with compressibility corrected
c. At mean sea level with standard conditions

98. An air data computer converts the indicated airspeed into the calibrated airspeed. Which two errors does it remove? ●

a. The error from the indicator and the error from the static pressure source
b. The error from the indicator and the error from air compressibility
c. The error from the pitot source and the error from air density

99. A mechanical standby airspeed indicator with bellows is fed with total and static pressure. Which airspeed does it display? ●

a. The equivalent airspeed
b. The calibrated airspeed
c. The indicated airspeed

100. Of the two speeds derived after the calibrated airspeed, the one needed for navigation and the one kept for design data are ●

a. the true airspeed and the indicated airspeed respectively
b. the true airspeed and the equivalent airspeed respectively
c. the equivalent airspeed and the true airspeed respectively

101. Under which condition is the speed formula valid ●

a. Only when the medium used is above the tropopause
b. Only when the medium used is incompressible
c. Only when the medium used is at standard temperature

102. An aircraft accelerates to a high speed and the total pressure rises more than the speed formula predicts. What produces this additional pressure? ●

a. Air expands at higher speeds and generates a total pressure decrease
b. Air compresses at lower speeds and generates a static pressure increase
c. Air compresses at higher speeds and generates a total pressure increase

103. How does the compressibility error change with airspeed and altitude ●

a. The higher the airspeed and altitude, the smaller the error
b. The lower the airspeed and the higher the altitude, the larger the error
c. The higher the airspeed and altitude, the larger the error

104. What is the equivalent airspeed ●

a. The true airspeed after the density calculation has been removed
b. The calibrated airspeed after the compressibility error has been removed
c. The indicated airspeed after the instrument error has been removed

105. Why is the equivalent airspeed not shown to the crew of a commercial airplane ●

a. It is only valid below the tropopause in standard conditions
b. It is only available when the air data computer is in test
c. It is only important for design data and further calculations

106. A navigation computation needs the speed of the aircraft through the surrounding air mass. Which airspeed is this? ●

a. The calibrated airspeed
b. The equivalent airspeed
c. The true airspeed

107. The compressibility error for a flight condition is known in knots. What is done with this value to obtain the equivalent airspeed? ●

a. It is subtracted from the true airspeed
b. It is added to the calibrated airspeed
c. It is subtracted from the calibrated airspeed

108. In which order are the airspeeds derived from the pressure inputs

a. IAS, CAS, EAS, TAS
b. CAS, IAS, TAS, EAS
c. IAS, EAS, CAS, TAS

109. Which step is made between EAS and TAS ●

a. A static source correction
b. A density calculation
c. A compressibility correction

110. Which correction is applied between CAS and EAS ●

a. Correction of instrument error
b. Correction of static source error
c. Correction of compressibility error

111. The pressure sensing element of an aneroid barometer is ●

a. a vertical tube filled with a fluid
b. a small capsule made of thin metal
c. a curved tube with an oval cross section

112. What creates the negative pressure at the top of a liquid barometer tube

a. The fluid flowing out of the tube due to its own weight
b. The expansion of the glass tube with temperature
c. The ambient air pressure acting on the container

113. A Bourdon tube transducer is opened for inspection. Which two parts carry out the pressure conversion? ●

a. The straight tube and a position sensor
b. The curved tube and a drive coil
c. The curved tube and a position sensor

114. A Bourdon tube transducer is fitted with a synchro transmitter. Which quantity does this sensor convert into an electrical signal? ●

a. The position of the tube tip
b. The pressure inside the tube
c. The temperature of the tube wall

115. How is an applied pressure distributed inside a Bourdon tube

a. Equally throughout the whole tube
b. Only at the sealed tube tip
c. Only at the oval section of the tube

116. What is the amount of tube tip movement proportional to ●

a. The applied pressure
b. The tube temperature
c. The signal frequency

117. Of the two capsule error influences, the one that depends on whether the pressure was rising or falling is ●

a. the hysteresis
b. the effect of temperature
c. the elastic aftereffect

118. A diaphragm capsule delivers only a stroke of 4 to 5 mm with a small force. What must the transfer mechanism do with it? ●

a. Convert it into a damped rotation of the needle
b. Convert it into an amplified rotation of the needle
c. Convert it into an amplified stroke of the capsule

119. Which effect does friction have in the transfer mechanism of an aneroid instrument ●

a. It falsifies the measured result
b. It damps the capsule stroke
c. It raises the capsule hysteresis

120. How is friction in the indicator mechanism kept to a minimum ●

a. With suitable bearings and clean construction of all moving parts
b. With heavier levers and increased spring tension on all moving parts
c. With sealed bearings and grease filling of all moving parts

121. In the altimeter one further measure helps to reduce the effect of friction. Which measure is this? ●

a. Vibration of the instrument
b. Damping of the capsule
c. Balancing of the needle

122. Play in the gearing between the pinion and the tooth segment would falsify the indication. How is this play taken out? ●

a. By spring tension on both sides
b. By increased bearing clearance
c. By spring tension on one side

123. What is meant by the hysteresis of a membrane capsule ●

a. The curve with increasing pressure differs from the curve with increasing temperature
b. The curve with increasing pressure differs from the curve with decreasing pressure
c. The stroke lags behind the pressure when a constant pressure acts for a long time

124. What causes the elastic aftereffect of a capsule ●

a. A temperature change acting on the capsule mounting
b. A constant pressure acting on the capsule over a longer period of time
c. A rapidly changing pressure acting on the capsule during a climb

125. What is the result of the slow yielding of a capsule under constant pressure ●

a. A change of the residual pressure
b. A time lag in the stroke
c. A permanent offset of the needle

126. Friction in the transfer mechanism, hysteresis, elastic aftereffect and temperature all work against precise operation. Which equipment do these error influences affect? ●

a. All aneroid type instruments
b. Only metric scale altimeters
c. Only digital air data computers

LEVEL 3 — APPLY AND REASON

127. RVSM airspace lies above FL 290 and is therefore flown on flight levels. The altimeter setting used there is ●

a. the QNH of the destination airfield
b. the standard pressure of 1013.2 hPa
c. the QFE of the departure airfield

128. An aircraft has two independent altimeters, an autopilot that can maintain the altitude and a device for indicating when the predetermined height is left. The further item of equipment it needs for flight in RVSM airspace is ●

a. a radio altimeter with an altitude alerting function
b. a second independent air data computer
c. a transponder with altitude transmission

129. An aircraft can no longer hold its selected flight level unless the pilot flies manually. Which requirement for flight in RVSM airspace is no longer met? ●

a. An autopilot that can maintain the altitude
b. A device for indicating when leaving the predetermined height
c. A transponder with altitude transmission

130. A dent is found in the skin close to a static port of an RVSM approved aircraft. Which consequence must be expected?

a. A considerable influence on the radio altimeter indication
b. A considerable influence on the altitude indication
c. A considerable influence on the mach number only

131. An altimeter fails in cruise above FL 290 and the aircraft no longer meets the requirements that apply there. Which action is required? ●

a. A descent to the transition level
b. A change of the altimeter setting to QNH
c. A quick exit of RVSM airspace

132. A standby airspeed indicator with bellows and the airspeed readout of the glass cockpit are read at the same moment and differ. Which airspeeds are the two showing? ●

a. Calibrated airspeed on the bellows instrument and indicated airspeed on the glass cockpit
b. Indicated airspeed on the bellows instrument and calibrated airspeed on the glass cockpit
c. Indicated airspeed on the bellows instrument and true airspeed on the glass cockpit

133. Navigation needs the real speed of the aircraft measured against the surrounding air, but the cockpit shows only the airspeed indication. Which speed is needed and which value must be applied to obtain it? ●

a. The true airspeed, obtained by applying temperature as a corrective value
b. The true airspeed, obtained by applying static pressure as a corrective value
c. The equivalent airspeed, obtained by applying temperature as a corrective value

134. Why is the equivalent airspeed lower than the calibrated airspeed at high speed and altitude? ●

a. Because compression of the air raises the total pressure, so the indication is too high
b. Because reduced density lowers the total pressure, so the indication is too high
c. Because compression of the air lowers the static pressure, so the indication is too low

135. Two aircraft fly at the same calibrated airspeed, one at 20,000 ft and one at 40,000 ft. Which one has the larger compressibility error?

a. Both have the same error
b. The aircraft at 40,000 ft
c. The aircraft at 20,000 ft

136. The synchro transmitter of a Bourdon tube transducer is found unserviceable while the tube itself is sound. Which step of the pressure measurement is lost? ●

a. The conversion of the tube tip position into an electrical signal
b. The deformation of the curved tube by the applied pressure
c. The supply of static pressure to the curved tube

137. An altimeter needle must turn 51 times over its full range. Which demand does this place on the instrument?

a. The highest demands on electrical power and signal bandwidth
b. The highest demands on material and fine mechanical precision
c. The highest demands on case sealing and static line length

138. Which measures are used to characterize the properties of a capsule for its intended use ●

a. Special alloys, artificial ageing and particular waveforms
b. Special coatings, artificial ageing and particular bearing materials
c. Special alloys, natural ageing and particular residual pressures

139. Which parts make temperature compensation demanding over the range -70 to +50 degrees C ●

a. The capsule mounting and the flexible tubes in the static system
b. The capsule mounting and the variable levers in the indicator mechanism
c. The pinion bearings and the quick release connectors of the instrument

140. An altimeter is held at one steady pressure on the test bench for a long time and the indication slowly drifts. Which error influence explains this? ●

a. The hysteresis of the capsule
b. The backlash of the pinion gearing
c. The elastic aftereffect of the capsule

141. An altimeter reads slightly differently at the same test pressure depending on whether the pressure was raised or lowered to reach it. Which error influence is shown? ●

- **a.** Hysteresis of the diaphragm capsule
- **b.** Elastic aftereffect of the diaphragm capsule
- **c.** Friction in the transfer mechanism

142. Through which parts must the smallest pressure change act to deflect the needle exactly

- **a.** From the capsule via levers, shaft and cogwheel
- **b.** From the capsule via bellows, resonator and coil
- **c.** From the diaphragm via strain gauges, bridge and amplifier

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Answers — Pressure measuring devices

76. **a** Reduced vertical separation minimum halves the vertical spacing used up high. (L1, PDF p. 377)

77. **c** RVSM airspace starts above FL 290 and uses 1000 ft spacing. (L1, PDF p. 377)

78. **a** IAS stands for indicated airspeed. (L1, PDF p. 381)

79. **c** Glass cockpits with an ADC display CAS. (L1, PDF p. 381)

80. **b** TAS is the speed required for navigation. (L1, PDF p. 383)

81. **b** The Bourdon tube arrangement turns pressure into a proportional electrical output. (L1, PDF p. 387)

82. **a** The tube has an oval cross section in the unpressurised state. (L1, PDF p. 387)

83. **b** The capsule stroke is only 4 to 5 mm. (L1, PDF p. 393)

84. **b** More usable levels mean more aircraft fit into the same volume of airspace. (L2, PDF p. 377)

85. **a** Altimeters become less precise with height, so a bigger buffer was needed. (L2, PDF p. 377)

86. **c** The list of RVSM-sensitive components closes with the structure and surroundings of the pitot-static sensors. (L2, PDF p. 377)

87. **c** The transponder must report altitude taken from the altitude measurement unit. (L2, PDF p. 377)

88. **c** More available levels give better routings, saving time and fuel. (L2, PDF p. 377)

89. **a** Improved precision of test equipment is the listed subject that deals with the equipment used to test the aircraft. (L2, PDF p. 379)

90. **b** Approval is per aircraft and is applied for by the operator. (L2, PDF p. 379)

91. **a** Each aircraft has to be modified in accordance with the required service bulletins and is subject to the altitude monitoring programme. (L2, PDF p. 379)

92. **b** Those three are the surface waviness causes listed. (L2, PDF p. 379)

93. **b** The training list names the part numbers of the air data computer. (L2, PDF p. 379)

94. **c** Airworthiness approval comes first and opens the way to the test flight. (L2, PDF p. 379)

95. **c** Operational approval is the last step and permits use of RVSM airspace. (L2, PDF p. 379)

96. **c** IAS follows dynamic pressure, which is total pressure minus static pressure. (L2, PDF p. 381)

97. **c** The bellows instrument is only true at mean sea level in standard conditions. (L2, PDF p. 381)

98. **a** CAS is IAS corrected for instrument error and static source error. (L2, PDF p. 381)

99. **c** A bellows instrument with no computer correction shows IAS. (L2, PDF p. 381)

100. **b** The true airspeed is the one wanted for navigation; the equivalent airspeed is not indicated and serves design data and further calculations. (L2, PDF p. 383)

101. **b** The formula assumes an incompressible medium. (L2, PDF p. 383)

102. **c** Compression of the air at high speed raises the total pressure, which the instrument reads as extra speed. (L2, PDF p. 383)

103. **c** The error grows with both airspeed and altitude. (L2, PDF p. 383)

104. **b** EAS is what is left when the compressibility error is taken out of CAS. (L2, PDF p. 383)

105. c EAS is a design and calculation value, not a cockpit indication. (L2, PDF p. 383)

106. c TAS is the aircraft speed relative to the surrounding air mass. (L2, PDF p. 383)

107. c Compression raises the total pressure, so the calibrated airspeed reads too high and the compressibility error has to be taken off it to leave the equivalent airspeed. (L2, PDF p. 383)

108. a The summary chain runs IAS to CAS to EAS to TAS. (L2, PDF p. 384)

109. b The last step of the chain is the density calculation. (L2, PDF p. 384)

110. c CAS becomes EAS when the compressibility error is corrected. (L2, PDF p. 384)

111. b The aneroid barometer senses pressure with a thin metal capsule; the fluid filled vertical tube belongs to the liquid barometer and the oval curved tube to the Bourdon transducer. (L2, PDF p. 385)

112. a The weight of the fluid draws it down and leaves a negative pressure above it. (L2, PDF p. 385)

113. c Only the curved oval tube and a position sensor at its tip are required. (L2, PDF p. 387)

114. a The sensor senses tip position, not pressure directly. (L2, PDF p. 387)

115. a The pressure acts equally over the whole tube. (L2, PDF p. 389)

116. a Tip travel is proportional to the applied pressure. (L2, PDF p. 389)

117. a Hysteresis is the difference between the rising and the falling pressure curve, while the elastic aftereffect comes from pressure held constant over a long time. (L2, PDF p. 393)

118. b The mechanism amplifies the tiny stroke into needle rotation. (L2, PDF p. 393)

119. a Friction in the mechanism corrupts the reading. (L2, PDF p. 393)

120. a Good bearings and clean mechanical construction minimise friction. (L2, PDF p. 393)

121. a Vibration keeps the altimeter mechanism from sticking, helping against friction. (L2, PDF p. 393)

122. c A one-sided spring tension takes up the gear backlash. (L2, PDF p. 393)

123. b Hysteresis is the difference between the rising and falling pressure characteristics. (L2, PDF p. 393)

124. b Long-held constant pressure makes the capsule yield slowly. (L2, PDF p. 393)

125. b The stroke lags in time, which is the elastic aftereffect. (L2, PDF p. 393)

126. a These errors act on all aneroid instruments. (L2, PDF p. 393)

127. b RVSM airspace is flight level airspace, and flight levels are flown with the standard pressure of 1013.2 hPa set. (L3, PDF p. 377)

128. c The RVSM equipment list is two independent altimeters, a device indicating departure from the predetermined height, an autopilot able to maintain the selected level and a transponder with altitude transmission connected to the altitude measurement unit; the transponder is the item still missing. (L3, PDF p. 377)

129. a One of the listed items of RVSM equipment is an autopilot able to maintain the altitude, and that is the function lost. (L3, PDF p. 377)

130. b Skin damage near the static source disturbs the static pressure and so the altitude reading. (L3, PDF p. 377)

131. c Losing RVSM capability means the aircraft must leave RVSM airspace quickly. (L3, PDF p. 379)

132. b A bellows instrument shows the indicated airspeed, while the glass cockpit indication is the calibrated airspeed. (L3, PDF p. 381)

133. a Navigation needs the true airspeed, and temperature is the corrective value applied so that the true airspeed can be indicated. (L3, PDF p. 383)

134. a The extra total pressure from compression makes the reading too high, so the correction reduces the speed. (L3, PDF p. 383)

135. b The compressibility error grows with altitude, so at the same calibrated airspeed the higher aircraft has the larger error. (L3, PDF p. 383)

136. a The synchro transmitter is the position sensor, so with it unserviceable the tube tip still moves but its position is no longer turned into an electrical signal. (L3, PDF p. 387)

137. b Such fine resolution needs the best material and fine mechanics. (L3, PDF p. 391)

138. a The three listed measures are special alloys, artificial ageing and particular waveforms. (L3, PDF p. 393)

139. b Compensation is hard because of the capsule mounting and the variable steering levers. (L3, PDF p. 393)

140. c Slow drift under a constant held pressure is the elastic aftereffect, not hysteresis. (L3, PDF p. 393)

141. a A difference between the rising and falling approach to the same pressure is hysteresis. (L3, PDF p. 393)

142. a The mechanical chain runs capsule, levers, shaft, cogwheel, needle. (L3, PDF p. 393)

Pressure transducers

PDF pages 395–401 · 23 questions

LEVEL 1 — RECALL

143. What is the function of a pressure transducer ●

a. To convert measured pressures into electrical signals
b. To convert electrical signals into measured pressures
c. To convert measured pressures into mechanical needle movement

144. In which equipment are pressure transducers used ●

a. In inertial reference units
b. In air data computers
c. In flight control computers

145. Several electronic pressure sensor principles exist. Which one is described as the most commonly used? ●

a. The force compensated cell
b. The piezoresistive transducer
c. The vibrating transducer

146. From which material is the micromechanical pressure transducer chamber made ●

a. Pyrex material
b. Metal material
c. Silicon material

147. What is the size of the micromechanical pressure transducer structure ●

a. Only a few micrometres
b. Only a few millimetres
c. Only a few centimetres

LEVEL 2 — UNDERSTAND AND EXPLAIN

148. Pressure transducers are fitted in air data computers because the computer needs the measured pressures ●

a. as an output frequency of the indicator
b. as amplified needle rotations
c. as electrical signals

149. Pressure transducers have converted the measured pressures into electrical signals. What happens to these signals next?

a. They are sent to the probe heater and transformed into the required current
b. They are sent to the computer and transformed into the required outputs
c. They are sent to the indicator and transformed into the required needle deflection

150. Which output does a vibrating transducer generate ●

a. An output frequency which depends on the input pressure
b. An output voltage which depends on the input pressure
c. An output frequency which depends on the reference pressure

151. Which function does the outer case of a vibrating transducer have

a. It forms the outer wall of the reference pressure chamber
b. It forms the outer wall of the measured pressure chamber
c. It forms the inner wall of the reference pressure chamber

152. What happens when pressure acts on the silicon lid of a piezoresistive transducer

a. The lid is dented and the resistance in its surface changes
b. The lid is dented and the capacitance in its surface changes
c. The lid vibrates and the frequency of its surface changes

153. What is the output voltage of a piezoresistive transducer proportional to ●

a. The input pressure
b. The supply frequency
c. The reference pressure

LEVEL 3 — APPLY AND REASON

154. A force compensated pressure cell is fitted instead of a plain pressure cell. Which two advantages are gained?

a. It is insensitive to pressure changes and has small instrument errors
b. It is sensitive to pressure changes and has small instrument errors
c. It is insensitive to pressure changes and needs no separate pressure cells

155. Which pressure sensing type needs no different pressure cells for static and pitot pressure?

a. The force compensated pressure cell
b. The plain pressure cell type
c. The vibration pressure transducer

156. An electronic pressure sensor delivers an output frequency that depends on the input pressure. This sensor is ●

a. the vibrating transducer, the most commonly used type
b. the piezoresistive transducer, the most commonly used type
c. the pressure cell transducer, the most commonly used type

157. One transducer gives an output frequency depending on the input pressure, another an output voltage in proportion to the input pressure. These are ●

a. the vibrating transducer and the micromechanical transducer respectively
b. the vibrating transducer and the Bourdon tube transducer respectively
c. the micromechanical transducer and the vibrating transducer respectively

158. A vibrating transducer is cut open. Where is the pressure that is being measured found?

a. In the reference chamber outside the vibrating cylinder
b. In the pressure chamber inside the vibrating cylinder
c. In the pressure chamber inside the center body

159. What is done with the signal from the pickup coil

a. It is digitized and used as reference, and fed back to the drive coil
b. It is amplified and used as output, and fed back to the reference chamber
c. It is amplified and used as output, and fed back to the drive coil

160. Why is the pickup coil signal fed back to the drive coil of a vibrating transducer?

a. To keep the cylinder oscillating at its natural frequency
b. To keep the reference chamber at a constant vacuum
c. To keep the output frequency at a constant value

161. A pressure sensor consists of a chamber made of silicon, only a few millimetres in size. It is ●

a. the micromechanical transducer, built as an aneroid-type chamber
b. the vibrating transducer, built as an aneroid-type chamber
c. the micromechanical transducer, built as a liquid-based chamber

162. Pressure bends the diaphragm of a piezoresistive transducer and the resistance of the strain gages changes. What does the bridge then deliver, and what is it proportional to?

a. An output voltage, in proportion to the supply voltage
b. An output voltage, in proportion to the input pressure
c. An output frequency, in proportion to the input pressure

163. Pressure is applied to a piezoresistive transducer. How do the two bridge voltages behave?

a. U1 decreases while U2 increases
b. U1 and U2 both increase
c. U1 increases while U2 decreases

164. A strain gauge wire is made thicker while its length stays the same. How does its resistance change?

a. It decreases with increasing area
b. It increases with increasing area
c. It is independent of the area

165. A strain gauge on the bent side of the diaphragm is stretched. Why does its resistance rise?

a. Because the wire becomes longer and resistance rises with length
b. Because the material constant of the wire rises with the strain
c. Because the wire becomes shorter and resistance rises as it shortens

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Answers — Pressure transducers

143. a Transducers turn pressure into electrical signals. (L1, PDF p. 395)

144. b Air data computers are one application of this transducer. (L1, PDF p. 395)

145. c The vibrating transducer is the most commonly used type. (L1, PDF p. 397)

146. c The chamber is made of silicon. (L1, PDF p. 399)

147. b The structure measures only a few millimetres. (L1, PDF p. 399)

148. c A transducer turns the measured pressures into electrical signals, which is the form the air data computer works with. (L2, PDF p. 395)

149. b The signals go to the computer, which makes the required outputs. (L2, PDF p. 395)

150. a Its output is a frequency that varies with the measured pressure. (L2, PDF p. 397)

151. a The outer case is the outer wall of the reference chamber. (L2, PDF p. 397)

152. a Denting the lid deforms the surface resistors and changes their resistance. (L2, PDF p. 399)

153. a The output voltage rises in proportion to the applied pressure. (L2, PDF p. 399)

154. a The force compensated cell is insensitive to pressure changes and keeps instrument errors small. (L3, PDF p. 396)

155. c The table gives 'no different pressure cells for static and pitot' for the vibration type; both pressure cell types are listed as needing different cells for static and pitot. (L3, PDF p. 396)

156. a The vibrating transducer is named as the most commonly used type and it is the one whose output frequency depends on the input pressure. (L3, PDF p. 397)

157. a The vibrating transducer produces a frequency; the micromechanical transducer produces a voltage in proportion to the pressure. (L3, PDF p. 397)

158. b The measured pressure sits in the chamber inside the cylinder. (L3, PDF p. 397)

159. c The pickup signal serves both as the amplified output and as the feedback to the drive coil. (L3, PDF p. 397)

160. a The feedback loop sustains vibration at the natural frequency. (L3, PDF p. 397)

161. a The micromechanical transducer is the aneroid-type chamber made of silicon whose structure is only a few millimetres in size. (L3, PDF p. 399)

162. b The four strain gages sit in a wheatstone bridge, and the change in their resistance gives an output voltage in proportion to the input pressure. (L3, PDF p. 401)

163. c The two halves of the bridge move in opposite directions when the plate bends. (L3, PDF p. 401)

164. a A larger cross section lowers the resistance. (L3, PDF p. 401)

165. a Resistance increases with the length of the wire, so stretching a strain gauge raises its resistance. (L3, PDF p. 401)

Pitot static systems

PDF pages 403–415 · 44 questions

LEVEL 1 — RECALL

166. An airspeed indicator in a simple system is connected to two pressure lines. Which pressures does it use? ●

a. Static pressure and differential pressure
b. Pitot pressure and total pressure
c. Pitot pressure and static pressure

167. What is static pressure ●

a. The actual atmospheric pressure surrounding the aircraft
b. The build-up of air pressure in a tube facing the airstream
c. The difference between total pressure and ram pressure

168. A pneumatic device is calibrated to the international standard atmosphere. Which value of air density at mean sea level applies? ●

a. 1,150 kg/m3
b. 1,013 kg/m3
c. 1,225 kg/m3

169. What does the abbreviation SSE stand for ●

a. Static system error
b. Standard source error
c. Static source error

170. What is pitot pressure ●

a. The build-up of air pressure in a tube with its opening across the airstream
b. The build-up of air pressure in a tube with its opening facing the airstream
c. The actual atmospheric pressure in the air surrounding the aircraft

171. Where is the pitot tube usually located on the aircraft ●

a. Near the tail, on the fin itself or below the rudder
b. Near the nose, on the wing leading edge or below the flaps
c. Near the nose, on the nose itself or below the cockpit

172. What does the pitot tube consist of and which pressure is present in it during the entire flight ●

a. A flapper plate chamber in which total pressure Ptot is present
b. A cylindrical pipe in which total pressure Ptot is present
c. A cylindrical pipe in which static pressure Ps is present

173. Why is the Prandtl tube electrically heated ●

a. To prevent condensation
b. To prevent icing
c. To prevent deformation

174. A Prandtl tube supplies both pressures from one probe. Where is it located on the aircraft? ●

a. At the rear of the aircraft
b. At the front of the aircraft
c. Under the wing of the aircraft

LEVEL 2 — UNDERSTAND AND EXPLAIN

175. What the altimeter and the vertical speed indicator have in common is that ●

a. Both use static pressure
b. Both use pitot pressure and static pressure
c. Both use pitot pressure

176. Compared with the simple system of a small aircraft, a modern commercial aircraft has

a. One pitot system feeding two static systems
b. One pitot system and one static system as well
c. Two or three separate pitot-static pressure systems

177. On which two quantities does air pressure p depend ●

a. Temperature and angle of attack
b. Air density and mach number
c. Air density and temperature

178. Besides the mach number, which flight condition value is listed as a factor the static source error depends on ●

a. The angle of attack
b. The outside air temperature
c. The air density

179. Why can a static source error arise on an aircraft ●

a. It is not always possible to calibrate the pneumatic devices to the ISA values
b. It is not always possible to find turbulence and ram air free zones on the aircraft
c. It is not always possible to keep the static ports electrically heated in flight

180. How large is the static source error on slow-flying propeller planes ●

a. So small that it must be measured
b. So large that it must be corrected
c. So small that it is negligible

181. The crew has to obtain the correct value from the distorted values being displayed. Where are the correction tables found?

a. In the Aircraft Maintenance Manual or in the flight log
b. In the Operating Manual or in the weight and balance data
c. In the Operating Manual or in the flight regulations

182. Which pressure alone is required to measure the speed of the aircraft in relation to the ambient air ●

a. The total pressure Ptot
b. The static pressure ps
c. The ram pressure q

183. Which statement correctly compares the electrical heating of the pitot tube with that of the static ports

a. The pitot tube is heated, the static ports only on some aircraft types
b. Only the static ports are heated, the pitot tube being self draining
c. Both are heated on every aircraft type

184. A pitot tube must be aligned in one particular direction when it is refitted because ●

a. the airstream keeps the pitot drain hole clear of water
b. pitot pressure builds up only in a tube facing the airstream
c. the airstream stops the pitot tube from freezing up

185. What do the small drain hole and the baffle of a pitot tube ensure ●

a. That ram air entering the tube is slowed down
b. That water entering the tube is drained away
c. That the tube inlet is kept free of any deformation

186. A student is worried that the drain hole in a pitot tube lets pressure escape. Which statement is correct? ●

a. The drain hole and the baffle slightly reduce the total pressure
b. The drain hole and the baffle do not affect the total pressure
c. The drain hole and the baffle slightly reduce the static pressure

187. What does the Prandtl tube combine in one unit ●

a. A pitot tube with a sensor to measure static pressure
b. A pitot tube with a sensor to measure total air temperature
c. A static port with a sensor to measure the angle of attack

188. A Prandtl tube is found with a rough surface after a repair. Which consequence must be expected?

a. It will lead to static pressure errors
b. It will lead to heating element failures
c. It will lead to total pressure errors

189. A Prandtl tube delivers total pressure and static pressure. Where is the required differential pressure obtained? ●

a. In the instrument or in a computer
b. In the tube itself or in the piping
c. In the static port or in the drain hole

LEVEL 3 — APPLY AND REASON

190. The pitot probe of a simple system becomes completely blocked while the static system stays clear. Which of the three supplied instruments is affected? ●

a. The altimeter
b. The airspeed indicator
c. The vertical speed indicator

191. During a repair the connection between the left and right hand measuring locations of one static pressure system is left open. What is lost by this? ●

a. The compensation for unbalance in pressure measurement
b. The heating of the static measuring locations
c. The supply of pitot pressure to the instruments

192. A technician traces one static pressure system of a small aeroplane and finds a measuring location on the left and on the right hand side of the fuselage. Which statement describes this arrangement correctly? ●

a. Both locations belong to the same pitot system and are connected together
b. Each location belongs to a separate static system and they stay isolated
c. Both locations belong to the same static system and are connected together

193. The Captain moves the handle of the alternate static source selector valve to its other position. Which static pressure then reaches his instruments? ●

a. The copilot's static pressure system
b. His own static pressure system
c. The pitot pressure of his own system

194. A helicopter cannot keep its static ports in air that is free of rotor downwash. Which fitting deals with this? ●

a. A drain hole at each static port
b. Special cover plates at the static ports
c. Electrically heated static ports

195. After a skin repair the zone around a static port is left with a rough surface. What must be expected at that port?

a. Pressure distorting turbulences
b. A loss of the port heating
c. A blockage of the pressure pipe

196. Why are static ports placed in areas with low turbulence and ram air distortion ●

a. Because turbulence and ram air distortion block the static pressure hoses
b. Because turbulence and ram air distortion overheat the static port zones
c. Because turbulence and ram air distortion falsify the static pressure reading

197. An aeroplane extends its flaps on the approach. Which reading can change as a result of this alone? ●

a. The static pressure reading
b. The angle of attack reading
c. The total air temperature reading

198. Of the factors the static source error depends on, the two that change with the configuration of the aircraft are ●

a. The mach number and the angle of attack
b. The mach number and the position of the flaps
c. The flap position and the nose gear position

199. An aeroplane changes its angle of attack in cruise with everything else unchanged. Which measurement may be affected by this? ●

a. The total pressure measurement
b. The static pressure measurement
c. The pitot pressure measurement

200. A slow-flying propeller plane and a jet aeroplane near the speed of sound differ above all in one of the static source error factors. Which factor is that? ●

a. The position of the flaps
b. The position of the nose gear
c. The mach number

201. A jet aeroplane flies near to the speed of sound and its static source error exceeds the tolerances required by law. Which correction becomes necessary? ●

a. A correction using an air data computer
b. A correction using a differential pressure bellows
c. A correction using an alternate static source valve

202. A static source error falsifies the static pressure supplied to a simple pitot-static system. Which instruments are affected? ●

a. Only the airspeed indicator of the system
b. Every instrument of the system
c. Only the vertical speed indicator of the system

203. In the relation Ptot = q + ps, the quantity ps is delivered to the instrument from ●

a. The pitot probe
b. The static ports
c. The drain hole of the pitot tube

204. An airspeed indicator is fed from the pitot probe as well as the static system because ●

a. The pitot probe cannot deliver a pressure without a static reference
b. The ram pressure it needs is the total pressure less the static pressure
c. The static pressure alone gives the speed in the surrounding air

205. A pitot tube delivers the pressure present in its cylindrical pipe throughout the flight. What must be supplied in addition before a speed can be obtained? ●

a. The static pressure from the static ports
b. A second total pressure from another pitot tube
c. The total air temperature from a temperature probe

206. An aircraft type is to take both the ram pressure and the static pressure needed for speed indication from one single ram air tube. Which probe is used? ●

a. A pitot tube
b. A multi function probe
c. A Prandtl tube

207. A Prandtl tube carries a feature that a plain pitot tube does not, namely ●

a. Holes drilled in the sides for the static lines
b. Electrical heating of the tube
c. A cylindrical inlet for the total pressure

208. Both a pitot tube and a Prandtl tube keep water out of the pitot line. Which item does this in the Prandtl tube together with the drain hole? ●

a. A baffle
b. A cover plate
c. A flapper plate

209. A Prandtl tube delivers Ptot and Ps to a computer. Which quantity does the computer obtain from them, and what is it needed for? ●

a. The ram pressure, needed for the speed indication
b. The total air temperature, needed for the speed indication
c. The static source error, needed for the altitude indication

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Answers — Pitot static systems

166. c The airspeed indicator is the only one of the three that needs both pitot and static pressure. (L1, PDF p. 403)

167. a Static pressure is simply the ambient atmospheric pressure around the aircraft. (L1, PDF p. 407)

168. c The ISA sea level density is 1,225 kg/m3. (L1, PDF p. 407)

169. c SSE is the static source error. (L1, PDF p. 409)

170. b Pitot pressure builds up in a tube whose opening points into the airstream. (L1, PDF p. 411)

171. c These are the three locations used. (L1, PDF p. 413)

172. b It is a cylindrical pipe carrying total pressure throughout the flight. (L1, PDF p. 413)

173. b Electrical heating keeps the probe free of ice. (L1, PDF p. 415)

174. b It sits at the front, like the pitot tube. (L1, PDF p. 415)

175. a Static pressure is named as the pressure used by each of these two instruments. (L2, PDF p. 403)

176. c The simple system has one of each, whereas the modern commercial aircraft has two or three separate systems. (L2, PDF p. 405)

177. c Air density and temperature are the two quantities involved. (L2, PDF p. 407)

178. a Angle of attack is one of the factors the static source error depends on, alongside the mach number and the two configuration items. (L2, PDF p. 409)

179. b Where no turbulence free and ram air free zone exists, the sensed static pressure is falsified. (L2, PDF p. 409)

180. c At low speed the error is negligible. (L2, PDF p. 409)

181. c The correction tables come from the Operating Manual and from the flight regulations. (L2, PDF p. 409)

182. c Only the dynamic or ram pressure q is needed for speed measurement. (L2, PDF p. 411)

183. a Electrical heating is standard for the pitot tube, whereas electrically heated static ports are fitted on some aircraft types only. (L2, PDF p. 413)

184. b Pitot pressure only builds up when the opening of the tube faces the airstream, so the alignment is what produces the pressure. (L2, PDF p. 413)

185. b Their job is to get rid of water that has entered the tube. (L2, PDF p. 413)

186. b The drain hole and baffle do not affect the total pressure. (L2, PDF p. 413)

187. a It puts the pitot function and a static pressure sensor into one probe. (L2, PDF p. 415)

188. a Roughness disturbs the flow over the side holes and falsifies static pressure. (L2, PDF p. 415)

189. a The subtraction is done in the instrument or in a computer, not in the probe. (L2, PDF p. 415)

190. b Of the three instruments only the airspeed indicator uses pitot pressure, so only it depends on the pitot probe. (L3, PDF p. 403)

191. a The two locations are joined only to compensate unbalance in pressure measurement, so breaking the joint removes that compensation. (L3, PDF p. 403)

192. c One static system uses both side locations and they are joined to each other. (L3, PDF p. 403)

193. a The valve exists so that the Captain can take his static pressure from the copilot's system instead of his own. (L3, PDF p. 405)

194. b The cover plates are fitted for exactly this purpose, to compensate the effects of the strong rotor downwash at the ports. (L3, PDF p. 407)

195. a The zone is required to be smooth precisely so that no pressure distorting turbulences occur, so a rough surface allows them. (L3, PDF p. 407)

196. c A port in a quiet zone senses the true static pressure, while turbulence and ram air distortion at the port falsify that reading. (L3, PDF p. 407)

197. a The static source error depends on the position of the flaps, and it is the static pressure reading that this error affects. (L3, PDF p. 409)

198. c Flaps and nose gear are the two listed items that are set by the aircraft configuration, the other two being flight condition values. (L3, PDF p. 409)

199. b The angle of attack is one of the factors the static source error depends on, and that error affects the static pressure reading. (L3, PDF p. 409)

200. c The two cases are set apart by how near the speed of sound they fly, which is the mach number among the listed factors. (L3, PDF p. 409)

201. a An air data computer correction is called for in this case. (L3, PDF p. 409)

202. b All three instruments of the simple system are supplied with static pressure, so all three are affected. (L3, PDF p. 409)

203. b The static pressure ps used in the relation is the pressure measured at the static ports. (L3, PDF p. 411)

204. b Speed needs the ram pressure, and the ram pressure is obtained by subtracting the static pressure from the total pressure delivered by the pitot probe. (L3, PDF p. 411)

205. a The pitot tube gives total pressure only, and the static pressure from the static ports has to be subtracted from it to get the ram pressure. (L3, PDF p. 413)

206. c The Prandtl tube is the single ram air tube that supplies both of these pressures. (L3, PDF p. 415)

207. a Both probes have a cylindrical pipe and electrical heating, but the side holes feeding static lines belong to the Prandtl tube. (L3, PDF p. 415)

208. c The Prandtl tube works with a flapper plate and a drain hole, the baffle being the item named for the pitot tube. (L3, PDF p. 415)

209. a The differential of the two delivered pressures is the ram pressure q, which is what the speed indication requires. (L3, PDF p. 415)

Pitot static system leaks

PDF pages 417–429 · 26 questions

LEVEL 1 — RECALL

210. An air data system receives three parameters from one single multi function probe. Which parameters are they? ●

a. Angle of attack, total air temperature and total pressure
b. Angle of attack, static air temperature and static pressure
c. Angle of sideslip, total air temperature and total pressure

211. Which part of the multi function probe measures the angle of attack ●

a. A TAT tube
b. A pitot probe
c. A wind vane

212. Which colour is used to designate pitot system connections ●

a. Red
b. Blue
c. Yellow

213. Two lines are parted at a quick-disconnect during maintenance. What happens at the coupling? ●

a. It vents the line to ambient air
b. It seals itself automatically
c. It stays open until a cap is fitted

214. A technician must check the mouth of a static port against its limits. Where are these limits normally found? ●

a. In ATA chapter 31
b. In ATA chapter 21
c. In ATA chapter 34

215. Which system controls the cabin rate ●

a. The anti-ice system
b. The airconditioning system
c. The pitot static system

LEVEL 2 — UNDERSTAND AND EXPLAIN

216. A multi function sensor must deliver the total pressure to the air data system. Which of its parts provides this parameter? ●

a. The temperature tube
b. The wind vane
c. The pitot probe

217. Each multi function probe has to pass its measured data to the related computer. Through which item does it do so?

a. Through the air data computer
b. Through the digital converter
c. Through the difference bellows

218. In some aircraft types the two pneumatic back up instruments are arranged so that

a. Both functions are integrated in one instrument
b. Both functions are shared between three instruments
c. Only the altitude function is retained

219. An instrument has been changed in the pitot static system. What must be carried out? ●

a. An approved leak-proof test of the system
b. An approved flushing of the static system
c. An approved accuracy check of the static ports

220. Why are a scale and a feeler-gage used at a static port ●

a. To check that the static port zone is smooth enough to prevent turbulences
b. To measure the mouth of the static port against the limits
c. To assess damage of the aircraft skin against the limits

221. Why must the static port zones have a smooth surface ●

a. So that no turbulences can occur
b. So that no ice can form on the port
c. So that water can drain away from the port

222. Where are the limits for damage of the skin at the static ports found ●

a. In the ATA chapter for the instruments or the IPC
b. In the ATA chapter for the aircraft structure or the SRM
c. In the ATA chapter for air conditioning or the wiring manual

223. What do the results of a leakage depend on

a. The size of the leakage and the altitude of the aircraft
b. The size of the leakage and whether the cabin is pressurized
c. The position of the leakage and the outside air temperature

224. A static line leaks inside the unpressurized cabin of an aircraft in flight. What does the altimeter do? ●

a. It freezes at the altitude flown when the leak started
b. It slightly decreases the indicated altitude
c. It slightly increases the indicated altitude

225. Which effect does a static system leakage inside an unpressurized cabin have on the airspeed indicator ●

a. It shows zero indicated airspeed
b. It shows a slightly higher indicated airspeed
c. It shows a slightly lower indicated airspeed

226. What happens to the pressure in the static lines when a static system leakage goes into a pressurized cabin ●

a. The pressure increases
b. The pressure stays unchanged
c. The pressure decreases

227. What happens to the indications of the altimeter and the airspeed indicator after a static leakage into a pressurized cabin ●

a. Both increase
b. Both stay unchanged
c. Both decrease

LEVEL 3 — APPLY AND REASON

228. An air data system is to receive the angle of attack and the total pressure from one single sensor. Which type is used? ●

a. A Prandtl tube
b. A pitot tube
c. A multi function probe

229. A multi function probe delivers Ptot to the air data system, so the quantity still required for a speed indication is ●

a. The total air temperature
b. The angle of attack
c. The static pressure

230. An aircraft with a modern EFIS carries two pneumatic back up instruments. Which of them has to be connected to the pitot system? ●

a. The standby altimeter
b. The standby airspeed indicator
c. Both of them equally

231. A line was parted only at a quick-disconnect. What applies to the leak-proof test?

a. It is required, but only on the static side
b. It is not required in this case
c. It is required, as after any other parting

232. A static line breaks inside an unpressurized cabin during flight. How does the vertical speed indicator behave? ●

a. It slowly drifts into a permanent climb indication
b. It jumps once and then stays at zero
c. It jumps once and then shows correct data

233. The same static line leak is considered in an unpressurized aircraft and in a pressurized one. How does the altimeter behave in each case? ●

a. It decreases the indication in the unpressurized aircraft and slightly increases it in the pressurized one
b. It slightly increases the indication in the unpressurized aircraft and decreases it in the pressurized one
c. It slightly increases the indication in both aircraft

234. Which behaviour does the vertical speed indicator show after a static system leakage in an unpressurized cabin compared with a pressurized cabin ●

a. An indication depending on cabin rate in the unpressurized cabin, and a small jump and then correct data in the pressurized cabin
b. A small jump and then correct data in both cabins
c. A small jump and then correct data in the unpressurized cabin, and an indication depending on cabin rate in the pressurized cabin

235. The static system of a pressurized aircraft is leaking inside. What does the vertical speed indication now depend on? ●

a. The aircraft rate of climb
b. The size of the leak
c. The cabin rate

Answers — Pitot static system leaks

210. a AOA, TAT and total pressure are the three outputs of the probe. (L1, PDF p. 417)

211. c The wind vane is the AOA sensing element. (L1, PDF p. 417)

212. a Pitot connections carry the red designation; yellow belongs to the static side. (L1, PDF p. 423)

213. b A self-sealing coupling closes the line itself the moment it is parted. (L1, PDF p. 423)

214. c The limits for the static port mouth are given in ATA chapter 34. (L1, PDF p. 425)

215. b The cabin rate is controlled by the airconditioning system. (L1, PDF p. 429)

216. c The built-in pitot probe supplies the total pressure. (L2, PDF p. 417)

217. b The digital converter is the link between the probe and the related computer. (L2, PDF p. 417)

218. a The two standby functions can be brought together in a single instrument on some types. (L2, PDF p. 421)

219. a Changing an instrument or a device calls for an approved leak-proof test. (L2, PDF p. 423)

220. c The scale and feeler-gage are the tools named for assessing skin damage against the AMM limits. (L2, PDF p. 425)

221. a A smooth zone keeps the airflow undisturbed so the port senses true static pressure. (L2, PDF p. 425)

222. b Skin damage is a structural item, so the structure chapter or the SRM holds those limits. (L2, PDF p. 425)

223. b Size of the leak and the pressurization state of the cabin are the two factors given. (L2, PDF p. 427)

224. c The lower cabin pressure reaching the static line reads as a slightly greater altitude. (L2, PDF p. 427)

225. b The airspeed indicator reads slightly high, in the same sense as the altimeter. (L2, PDF p. 427)

226. a Cabin pressure is above the ambient static pressure, so the static lines see a rise. (L2, PDF p. 429)

227. c The raised static pressure drives both indications down. (L2, PDF p. 429)

228. c Only the multi function probe gives angle of attack as well as total pressure. (L3, PDF p. 417)

229. c The ram pressure needed for speed is the total pressure less the static pressure, and the probe supplies only the total pressure of the two. (L3, PDF p. 417)

230. b An airspeed indicator uses pitot pressure as well as static pressure, while an altimeter uses static pressure only. (L3, PDF p. 421)

231. b A parting at a quick-disconnect is the one exception to the leak-proof test requirement. (L3, PDF p. 423)

232. c The VSI reacts only to the sudden change and then works normally again. (L3, PDF p. 427)

233. b With an unpressurized cabin the leak lowers the pressure in the static lines, so the altimeter reads slightly higher, while cabin pressure entering the lines of a pressurized aircraft raises that pressure, so it reads lower. (L3, PDF p. 429)

234. c The unpressurized case gives a single small jump and then correct data, while in the pressurized aircraft the indication follows the cabin rate. (L3, PDF p. 429)

235. c With cabin air in the static line the VSI follows the cabin rate, not the aircraft. (L3, PDF p. 429)

Pitot static blockages and icing

PDF pages 431–443 · 30 questions

LEVEL 1 — RECALL

236. Which instrument is affected by a leakage in the pitot system ●

a. The vertical speed indicator only
b. The airspeed indicator only
c. The altimeter only

237. What does a completely blocked pitot tube mean ●

a. The drain hole is blocked and the inlet is open
b. The inlet is blocked and the drain hole is open
c. The inlet and the drain hole are blocked

238. A pitot tube is completely blocked by ice. Which instruments are not affected? ●

a. The altimeter and the airspeed indicator
b. The altimeter and the vertical speed indicator
c. The airspeed indicator and the vertical speed indicator

239. The indications fluctuate during a check of the pitot static system. What does this show? ●

a. The presence of water in the pressure pipes
b. The presence of ice at the static port
c. The presence of a leak in the pressure pipes

LEVEL 2 — UNDERSTAND AND EXPLAIN

240. A pitot line leaks inside an unpressurized cabin. What does the airspeed indication do? ●

a. It increases into the overspeed region
b. It decreases until it reaches zero
c. It stays at the value flown when the leak started

241. Why does the airspeed indication decrease when both the pitot and the static system leak ●

a. The total pressure in the pitot line is trapped at a constant value
b. Pitot and static pressure in the lines tend to be the same
c. The static pressure in the lines rises above the pitot pressure

242. Both the pitot system and the static system are leaking. What do the altimeter and the vertical speed indicator show? ●

a. The same results as for a leakage in the pitot system
b. The same results as for a leakage in the static system
c. Correct indications, because the two leaks cancel each other

243. What happens to the indications with a blocked static system while the aircraft flies at constant altitude and constant airspeed ●

a. The airspeed indication slowly decreases
b. The altitude indication slowly increases
c. There are no changes in the indication

244. Which indications do the altimeter and the vertical speed indicator give with a blocked static system, during a climb and during a descent ●

a. The same altitude and a vertical speed that follows the aircraft
b. A changing altitude and a vertical speed of zero
c. The same altitude and a vertical speed of zero

245. Why does the airspeed indicator react like an altimeter when the pitot tube is totally blocked ●

a. The total pressure is trapped, so only the changing static pressure moves the indication
b. Total and static pressure become nearly equal, so the indication falls away
c. The static pressure is trapped, so only the changing total pressure moves the indication

246. What happens to the pitot pressure when the pitot tube inlet is blocked but the water drain hole is still open ●

a. It decreases to the value of the static pressure
b. It increases to the value of the total pressure
c. It stays at the value it had before the blockage

247. Why are the altimeter and the vertical speed indicator not affected by a blocked pitot inlet ●

a. They get both pitot and static pressure
b. They only get pitot pressure
c. They only get static pressure

248. Why can water be present in the pressure pipes ●

a. Their test fittings are used to simulate altitude and speed
b. They are not drained on a regular basis
c. They are capped for protection during testing

249. The static ports of a cold aeroplane are capped and the aircraft is moved into a warm hangar. What happens in the static system?

a. The pressure decreases
b. The pressure stays the same
c. The pressure increases

250. What can be said about the ram pressure when both the pitot inlet and the drain are iced over ●

a. It is always zero
b. It depends on the altitude
c. It is constant at every altitude

LEVEL 3 — APPLY AND REASON

251. Why does the airspeed indication fall to zero with a pitot leakage in an unpressurized cabin ●

a. Static and pitot pressure become nearly the same
b. The pitot pressure rises with the cabin pressure
c. The static pressure becomes much greater than the pitot pressure

252. A pitot line leaks inside a pressurized cabin. What can be said about the airspeed indication? ●

a. It always falls to zero, because pitot and static pressure become equal
b. It always rises, because the cabin pressure is higher than the static pressure
c. It is unpredictable, because it depends on altitude and cabin pressure

253. Which instruments are disturbed by a leakage in the pitot system alone, compared with a leakage in both the pitot and the static system ●

a. All three instruments, compared with the airspeed indicator alone
b. The airspeed indicator and the altimeter, compared with all three instruments
c. Only the airspeed indicator, compared with all three instruments

254. Both the pitot system and the static system leak inside an unpressurized cabin. What does the altimeter do? ●

a. It slightly increases the indicated altitude
b. It holds the altitude it had when the leakage started
c. It slightly decreases the indicated altitude

255. An aircraft cruises at constant altitude and constant airspeed with a blocked static system and then begins a climb. What does the altimeter do? ●

a. It increases its altitude indication as the climb starts
b. It continues to display the same altitude
c. It jumps once and then follows the climb correctly

256. Why does the airspeed indication decrease during a climb with a blocked static system ●

a. The total pressure falls while the trapped static pressure stays constant
b. The total pressure and the static pressure both decrease together
c. The total pressure from the pitot tube stays constant while the static pressure decreases

257. How does the airspeed indication behave with a blocked static system during a climb compared with a descent ●

a. Decreasing during both the climb and the descent
b. Increasing during the climb and decreasing during the descent
c. Decreasing during the climb and increasing during the descent

258. Why does the airspeed indication increase during a descent with a blocked static system ●

a. The total pressure and the static pressure both increase together
b. The total pressure rises while the trapped static pressure stays constant
c. The trapped static pressure increases while the total pressure remains constant

259. Two aircraft climb, one with a blocked static system and one with a completely blocked pitot tube. What does the airspeed indicator show in each? ●

a. A decreasing indication in both aircraft
b. A decreasing indication with the blocked static system and an increasing indication with the blocked pitot tube
c. An increasing indication with the blocked static system and a decreasing indication with the blocked pitot tube

260. How does the airspeed indication behave during a descent with a blocked static system compared with a completely blocked pitot tube ●

a. Decreasing with the blocked static system and increasing with the blocked pitot tube
b. Increasing with the blocked static system and decreasing with the blocked pitot tube
c. Increasing with both blockages

261. The pitot tube becomes totally blocked in cruise and the thrust is then increased to accelerate. What does the airspeed indicator show? ●

a. A rise in the indicated speed
b. A fall of the indicated speed to zero
c. No change in the indicated speed

262. The pitot inlet of an aircraft is found blocked by ice. Which indication shows that the water drain hole is still open? ●

a. The airspeed indication increases into the overspeed region
b. The airspeed indication stops responding to changes of thrust
c. The airspeed indication decreases all the way to zero

263. What happens to the pitot pressure and the airspeed indication when the pitot inlet is iced over but the drain is free ●

a. The pitot pressure remains constant and the airspeed indication decreases to zero
b. The pitot pressure falls to static pressure and the airspeed indication increases
c. The pitot pressure falls to static pressure and the airspeed indication decreases to zero

264. The pitot inlet is iced over but the drain is free. What does this tell you about the heating? ●

a. The heating is defective or deactivated
b. The heating is working correctly
c. The heating is insufficient

265. Both the pitot inlet and the drain are iced over. What does this say about the heating and the pitot pressure? ●

a. The heating is defective or deactivated and the pitot pressure remains constant
b. The heating is defective or deactivated and the pitot pressure falls to static pressure
c. The heating is insufficient and the pitot pressure falls to static pressure

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Answers — Pitot static blockages and icing

236. b Only the airspeed indicator is fed from the pitot system. (L1, PDF p. 431)

237. c Complete blockage covers both the inlet and the drain hole. (L1, PDF p. 439)

238. b Those two run on static pressure, which the pitot blockage does not touch. (L1, PDF p. 439)

239. a Fluctuating indications are the sign of water lying in the pipes. (L1, PDF p. 443)

240. b With the pitot line open to the cabin the airspeed indication falls away to zero. (L2, PDF p. 431)

241. b Both lines fill with the same cabin pressure, so the difference the indicator works on shrinks. (L2, PDF p. 433)

242. b Those two instruments see only static pressure, so the static leak result still applies. (L2, PDF p. 433)

243. c Nothing changes while the trapped static pressure still matches the real one. (L2, PDF p. 435)

244. c Both instruments work from the trapped static pressure, so the altitude stays frozen and the vertical speed stays at zero in the climb and in the descent alike. (L2, PDF p. 437)

245. a With the total pressure held fixed the indication is driven only by static pressure, which is what an altimeter responds to. (L2, PDF p. 439)

246. a The open drain lets the trapped pressure bleed down to static. (L2, PDF p. 441)

247. c Both instruments are fed from the static system alone. (L2, PDF p. 441)

248. b The pipes are not drained routinely, so water can collect in them. (L2, PDF p. 443)

249. c Warming the trapped air in a sealed static system raises its pressure. (L2, PDF p. 443)

250. b With the total pressure held fixed, only the static pressure varies, so ram pressure follows altitude. (L2, PDF p. 443)

251. a The airspeed indicator works on the difference between the two pressures, which is now almost nil. (L3, PDF p. 431)

252. c With a pressurized cabin the two varying pressures make the indication unpredictable. (L3, PDF p. 431)

253. c A leak in the pitot system alone affects the airspeed indicator only; once the static system leaks as well, the altimeter and the vertical speed indication are disturbed too. (L3, PDF p. 433)

254. a With both systems leaking the altimeter shows the same result as for a static leak, and in an unpressurized cabin that result is a slightly increased indicated altitude. (L3, PDF p. 433)

255. b The trapped static pressure cannot change, so the altimeter holds the same altitude in level flight and through the climb. (L3, PDF p. 435)

256. a Only the total pressure falls; the blocked static side cannot follow it. (L3, PDF p. 435)

257. c The trapped static pressure stays fixed while the total pressure falls in the climb and rises in the descent, so the indication first decreases and then increases. (L3, PDF p. 437)

258. b Only the total pressure rises; the blocked static side stays where it was. (L3, PDF p. 437)

259. b With the static side blocked the climb gives a decreasing indication, while with the pitot side blocked the total pressure is trapped and the falling static pressure drives the indication up. (L3, PDF p. 439)

260. b With the static side blocked the descent gives an increasing indication, while with the pitot side blocked the total pressure is trapped and the rising static pressure drives the indication down. (L3, PDF p. 439)

261. c At constant altitude neither pressure changes, so the reading is frozen whatever the thrust. (L3, PDF p. 439)

262. c With the drain open the pitot pressure falls to static and the indication drops to zero, which a completely blocked tube does not do. (L3, PDF p. 441)

263. c The open drain lets the pitot pressure fall to static pressure, leaving no difference for the indicator to show. (L3, PDF p. 443)

264. c Ice only at the inlet points to heating that is working but not enough. (L3, PDF p. 443)

265. a Complete icing means the heating has failed or is off, and the sealed pitot pressure cannot change. (L3, PDF p. 443)